DISCRETE MATH · SEQUENCES, MATHEMATICAL INDUCTION, AND RECURSION

Recursion & Closed Form Explorer

Explore & Reflect **TEACHER KEY**

Name _____ Date _____ Period _____

Open **Recursion & Closed Form Explorer** in your unit's Interactive Math section on Mr. K's Classroom (or use your teacher's link). Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3. Do Parts 4 and 5 on your own.

1 Predict before you touch the tool

1. $a_n = 2a_{n-1} + 1$ with $a_1 = 1$ (Tower of Hanoi). Predict a_5 and a_{10}.
   $a_5 = 31$, $a_{10} = 1023$.
2. Same rule, but $a_1 = 4$. Do you get the same sequence? Why does a recurrence need initial terms?
   No (4, 9, 19, 39, …). The rule only says how to get the next term; the initial term decides which sequence.

2 Explore use the tool

1. Arithmetic: set c = 1. Find d and a_1 so that $a_1 = 5$ and $a_{10} = 32$. Record the closed form.
   $d = 3$, $a_1 = 5$: $a_n = 5 + 3(n - 1) = 3n + 2$.
2. Tower of Hanoi: type a closed form that matches every term. What does the tool say about proof?
   $a_n = 2^n - 1$. Matching 15 terms is evidence, not a proof; prove it by induction.
3. Find c, d and a_1 with $c \neq 1$ that make a constant sequence. What is the relationship?
   $a_1 = d/(1 - c)$, the fixed point. Example: $c = 3$, $d = -10$, $a_1 = 5$.
4. Fibonacci ($a_1 = a_2 = 1$): record a_{12} and S_{10}. Compare S_{10} with a_{12}.
   $a_{12} = 144$, $S_{10} = 143 = a_{12} - 1$.

3 Record use the tool and fill in the table

Recurrences and closed forms

Enter each recurrence, then test your closed form in the guess box.

Recurrence and initial term	$a_1 \ldots a_5$	Type	Closed form
$a_n = a_{n-1} + 4$, $a_1 = 7$	7, 11, 15, 19, 23	arithmetic	$4n + 3$
$a_n = 3a_{n-1}$, $a_1 = 2$	2, 6, 18, 54, 162	geometric	$2 \cdot 3^{n-1}$
$a_n = 2a_{n-1} + 1$, $a_1 = 1$	1, 3, 7, 15, 31	neither	$2^n - 1$
$a_n = 0.5a_{n-1} + 3$, $a_1 = 2$	2, 4, 5, 5.5, 5.75	neither (approaches 6)	$6 - 4(0.5)^{n-1}$

4 Explain what did you discover?

1. If c = 1 the sequence is **arithmetic** with common difference d; if d = 0 it is **geometric** with common ratio c.
2. A closed form that matches many terms is a **conjecture** until it is proved.
3. Induction proof of $a_n = f(n)$: show $f(1) = a_1$ (the **basis** step), then assume $a_k = f(k)$ and show $c \cdot f(k) + d =$ **f(k + 1)**.

4. Prove by induction that $a_n = 2^n - 1$ for $a_n = 2a_{n-1} + 1$, $a_1 = 1$.
   Basis: $2^1 - 1 = 1 = a_1$. Inductive step: if $a_k = 2^k - 1$, then $a_{k+1} = 2a_k + 1 = 2(2^k - 1) + 1 = 2^{k+1} - 1$. So the formula holds for all $n \geq 1$.

5 Apply a new situation, no tool

1. $a_n = a_{n-1} - 6$, $a_1 = 50$. Find a_{20} with a closed form.
   $a_n = 50 - 6(n - 1)$; $a_{20} = 50 - 114 = -64$.
2. A \$1000 deposit grows 5% a year and \$100 is added each year after the first: $a_n = 1.05a_{n-1} + 100$, $a_1 = 1000$. Find a_3.
   $a_2 = 1150$, $a_3 = 1307.50$.
3. Find a closed form for $a_n = 3a_{n-1}$, $a_1 = 5$, and find a_6.
   $a_n = 5 \cdot 3^{n-1}$; $a_6 = 1215$.
4. Show that the guess $a_n = n^2 - n + 1$ matches $a_n = a_{n-1} + 2(n - 1)$, $a_1 = 1$ for $n = 1, 2, 3$, then do the inductive step.
   1, 3, 7 both ways. If $a_k = k^2 - k + 1$ then $a_{k+1} = k^2 - k + 1 + 2k = (k + 1)^2 - (k + 1) + 1$.

Teacher notes

Intended discovery: A recurrence plus initial terms determines a sequence; $c = 1$ and $d = 0$ give arithmetic and geometric sequences; first-order sequences approach or move away from the fixed point $d/(1 - c)$; a matching closed form is only a conjecture until proved by induction.

Likely misconceptions:

- Thinking agreement on several terms proves a formula.
- Forgetting the initial term (or using a_0 vs a_1 inconsistently).
- In the inductive step, assuming what must be shown (using $f(k + 1)$ instead of deriving it).

Answers that can vary: Explore 3 values vary (any $a_1 = d/(1 - c)$); equivalent closed forms ($3n + 2$ vs $5 + 3(n - 1)$) are all correct.